EETS 7302 TCP/IP Network Administration

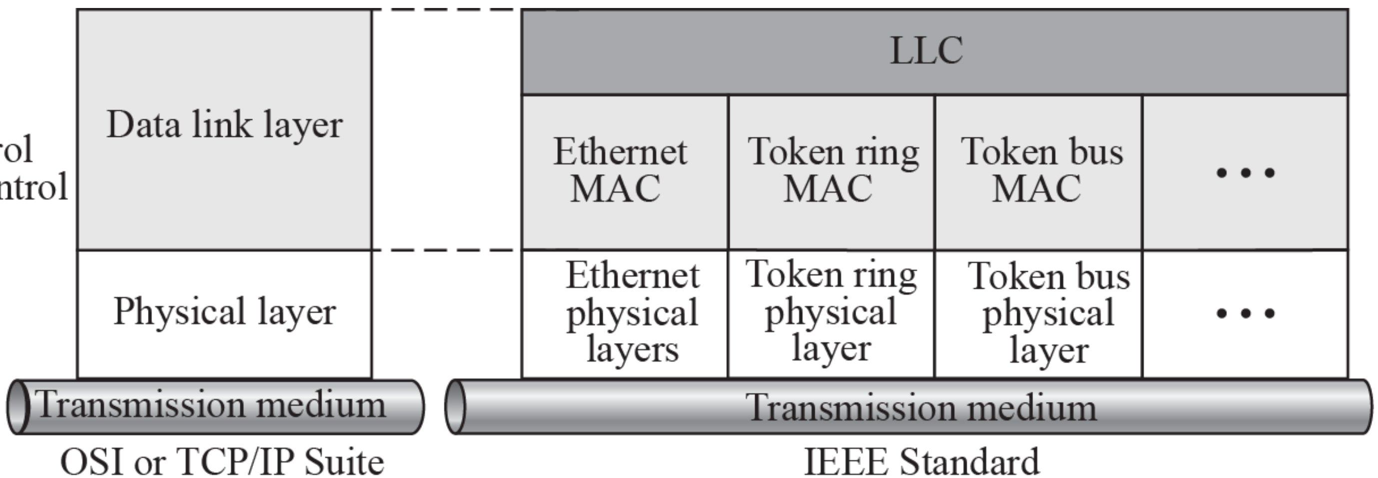
Session 2

LAN - LOCAL AREA NETWORKS

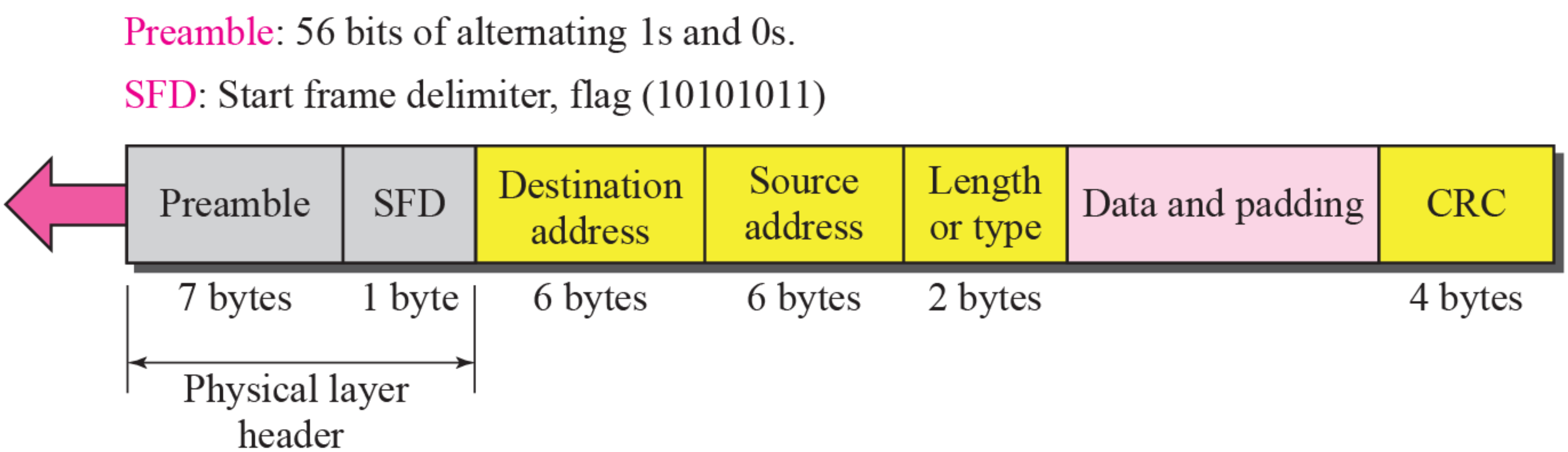
- **A local area network (LAN) is a computer network that is designed for a limited geographic area such as a building or a campus.**
- **Most LANs today are also linked to a wide area network (WAN) or the Internet.**
- **LAN market has seen several technologies such as Ethernet, token ring, token bus, FDDI, and ATM LAN, but Ethernet is by far the dominant technology.**

IEEE standard for LANs

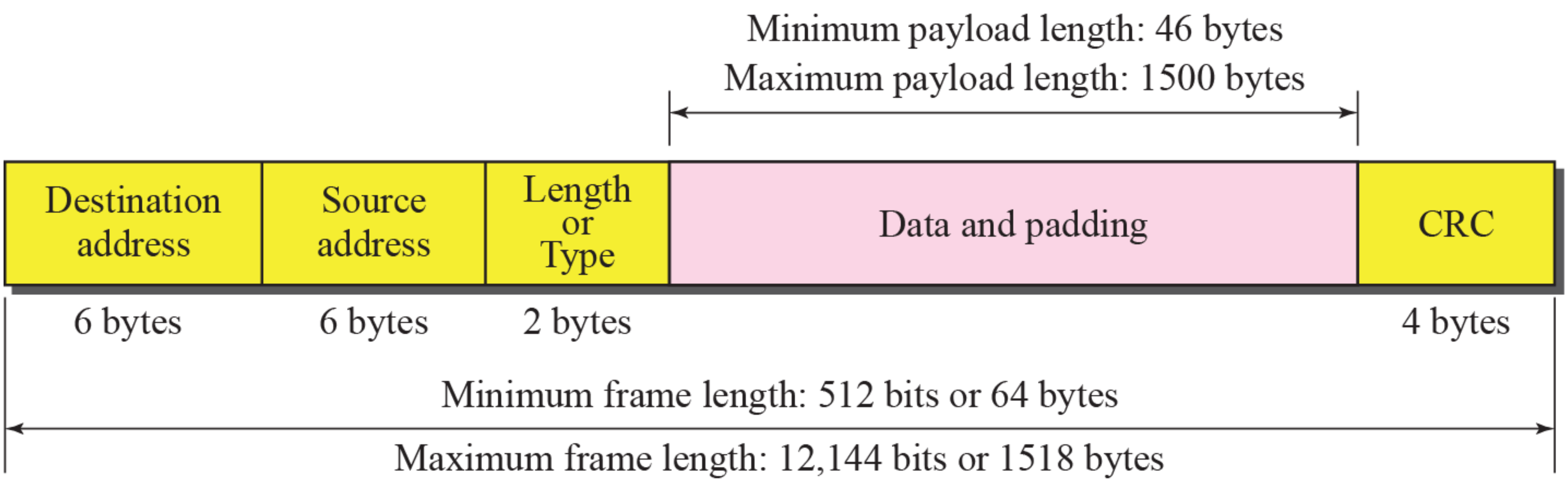
LLC: Logical link control
MAC: Media access control



Ethernet Frame



Maximum and minimum lengths



Minimum length: 64 bytes (512 bits)

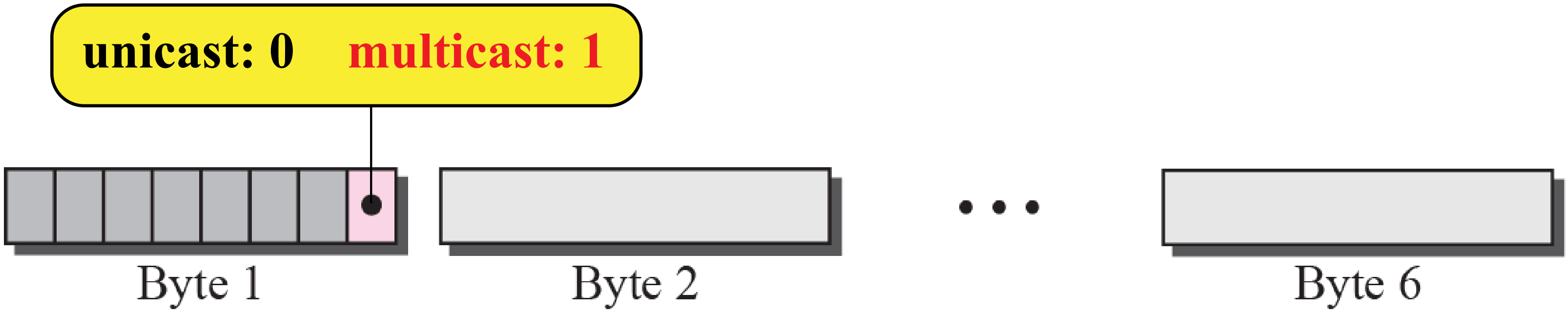
Maximum length: 1518 bytes (12,144 bits)

Ethernet address in hexadecimal notation

d: Hexadecimal digit

$$\boxed{d_1d_2 : d_3d_4 : d_5d_6 : d_7d_8 : d_9d_{10} : d_{11}d_{12}}$$

6 bytes = 12 hexadecimal digits = 48 bits



The broadcast destination address is a special case of the multicast address in which all bits are 1s.

The least significant bit of the first byte defines the type of address. If the bit is 0, the address is unicast; otherwise, it is multicast.

Example (5 min. exercise)

Define the type of the following destination addresses:

a. 4A:30:10:21:10:1A

b. 47:20:1B:2E:08:EE

c. FF:FF:FF:FF:FF:FF

Solution

?

?

?

Example

Define the type of the following destination addresses:

a. **4A:30:10:21:10:1A**
b. **47:20:1B:2E:08:EE**
c. **FF:FF:FF:FF:FF:FF**

Solution

To find the type of the address, we need to look at the second hexadecimal digit from the left. If it is even, the address is unicast. If it is odd, the address is multicast. If all digits are F's, the address is broadcast. Therefore, we have the following:

a. **This is a unicast address because A in binary is 1010 (even).**
b. **This is a multicast address because 7 in binary is 0111 (odd).**
c. **This is a broadcast address because all digits are F's.**

Example (5.min exercise)

Show how the address 47:20:1B:2E:08:EE is sent out on line.

Solution

?

Example

Show how the address 47:20:1B:2E:08:EE is sent out on line.

Solution

The address is sent left-to-right, byte by byte; for each byte, it is sent right-to-left, bit by bit, as shown below:

← 11100010 00000100 11011000 01110100 00010000 01110111

Ethernet evolution through four generations

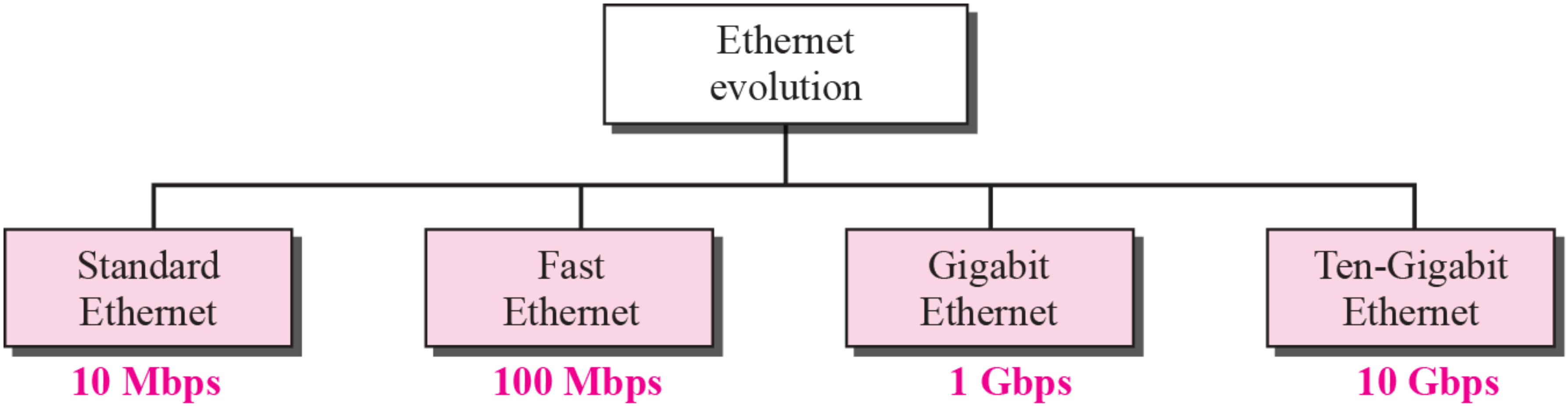
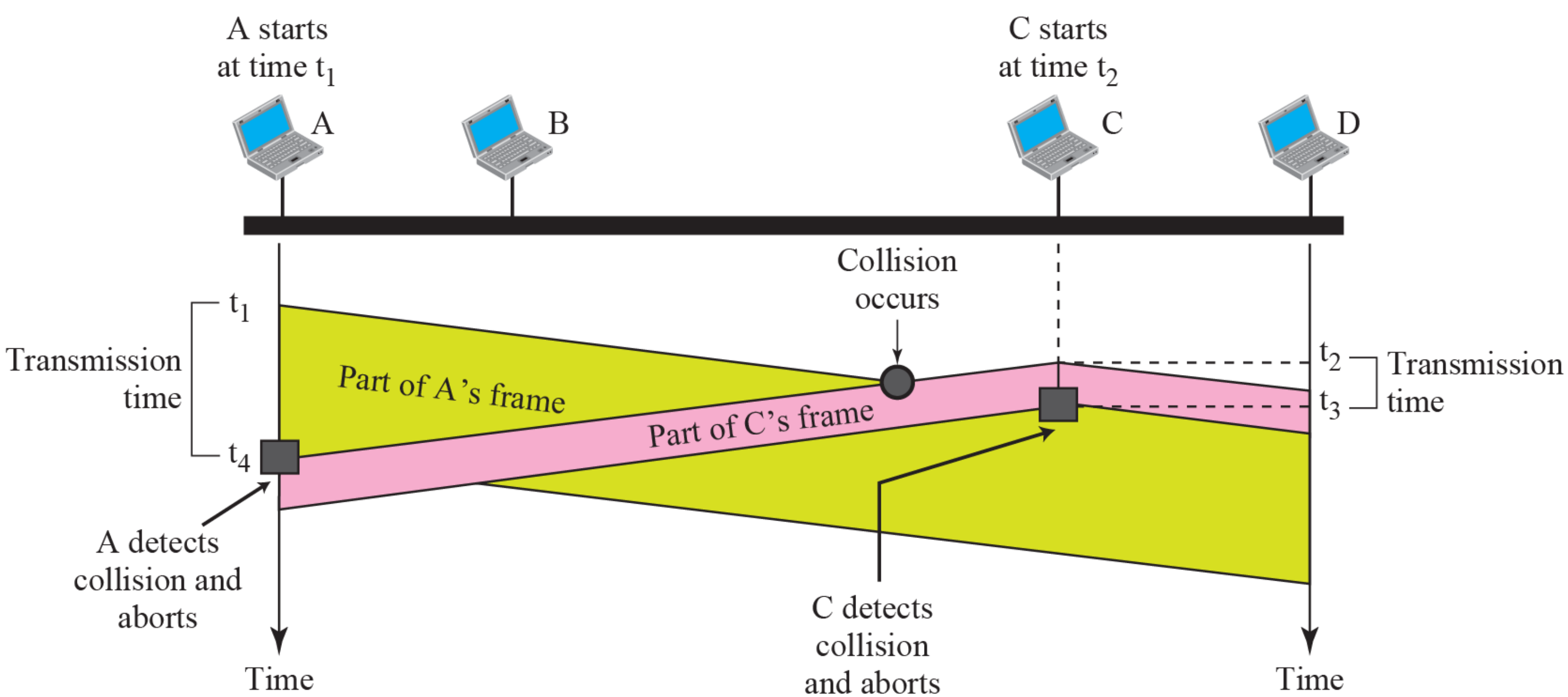
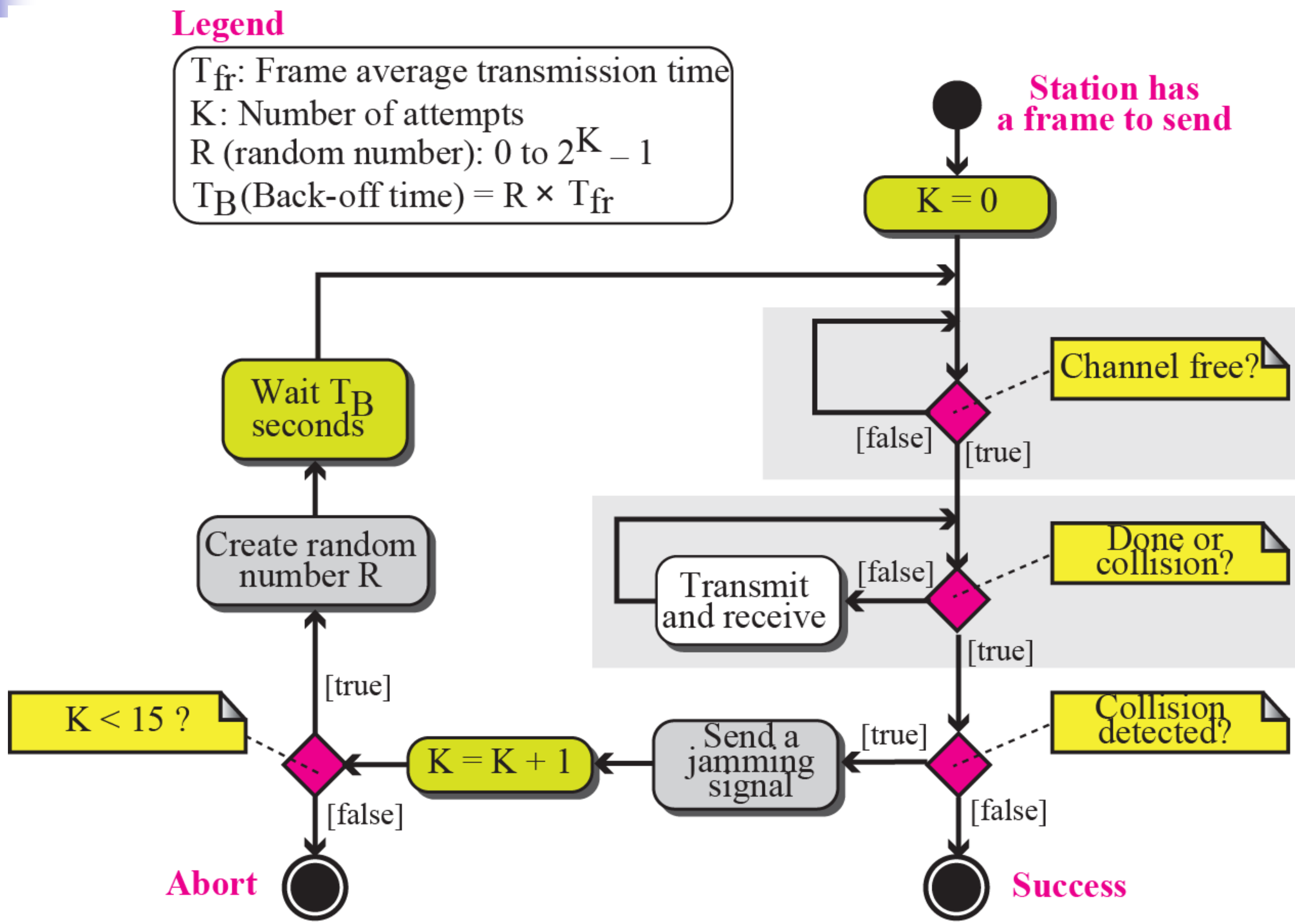


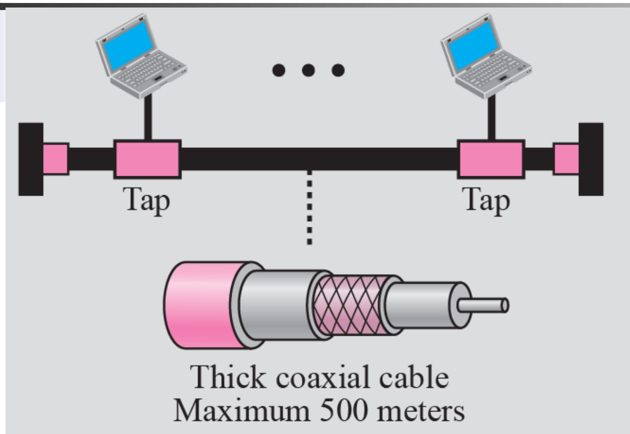
Figure 3.8 *Collision of the first bit in CSMA/CD*



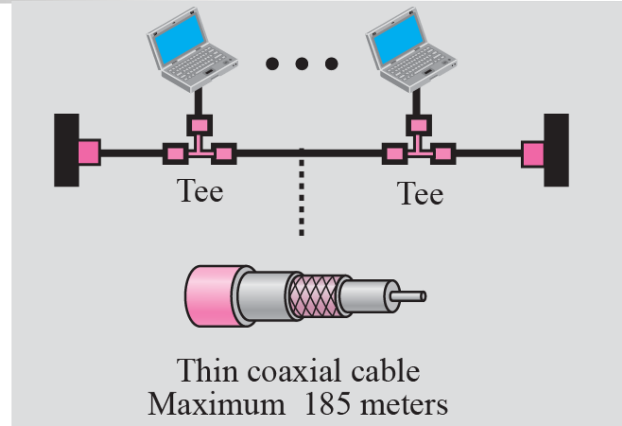
CSMA/CD flow diagram



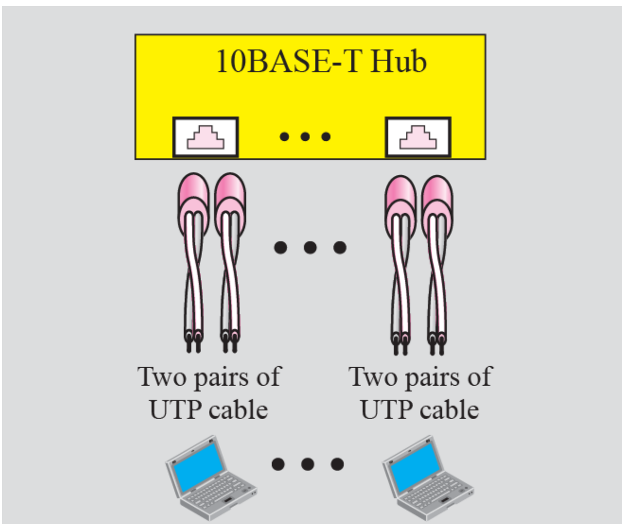
Standard Ethernet implementation



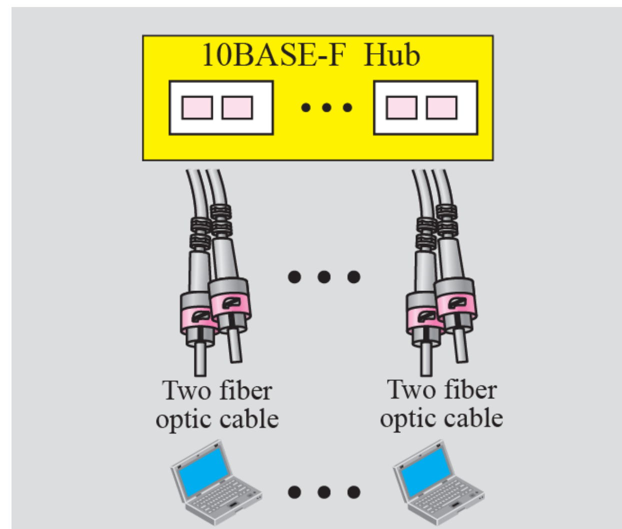
a. 10BASE5



b. 10BASE2



c. 10BASE-T

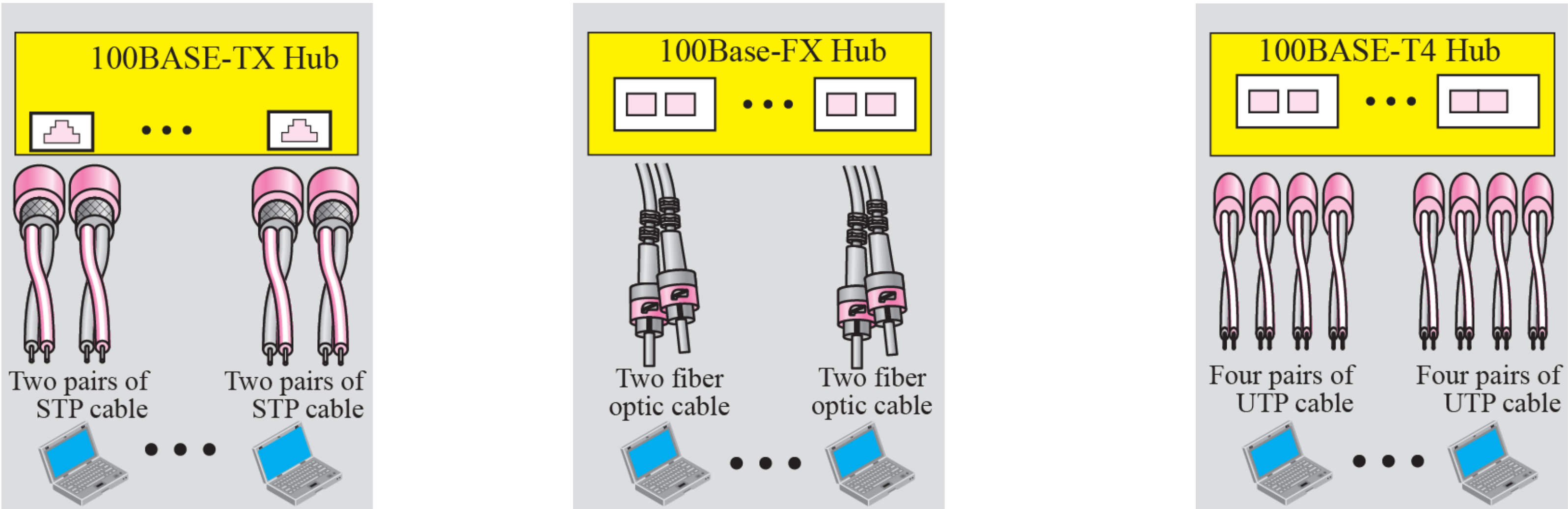


d. 10BASE-F

Table 3.1 *Summary of Standard Ethernet implementations*

Characteristics	*10Base5*	*10Base2*	*10Base-T*	*10Base-F*
Medium	Thick coax	Thin coax	2 UTP	2 Fiber
Maximum length	500 m	185 m	100 m	2000 m

Fast Ethernet implementation



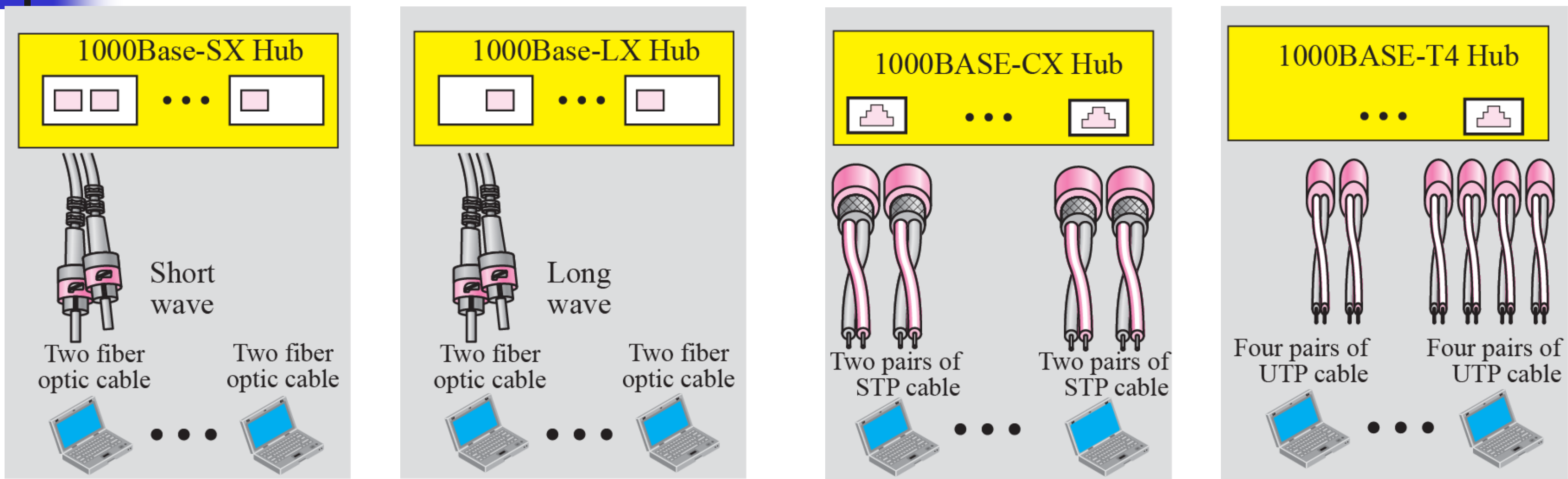
a. 100BASE-TX b. 100BASE-FX c. 100BASE-T4

Table 3.2 *Summary of Fast Ethernet implementations*

Characteristics	*100Base-TX*	*100Base-FX*	*100Base-T4*
Media	STP	Fiber	UTP
Number of wires	2	2	4
Maximum length	100 m	100 m	100 m

STP: Shielded Twisted Pair
UTP: Unshielded Twisted Pair

Gigabit Ethernet implementation



a. 1000BASE-SX b. 100BASE-LX c. 1000BASE-CX d. 1000BASE-T4

Table 3.3 *Summary of Gigabit Ethernet implementations*

Characteristics	*1000Base-SX*	*1000Base-LX*	*1000Base-CX*	*1000Base-T4*
Media	Fiber short-wave	Fiber long-wave	STP	Cat 5 UTP
Number of wires	2	2	2	4
Maximum length	550 m	5000 m	25 m	100 m

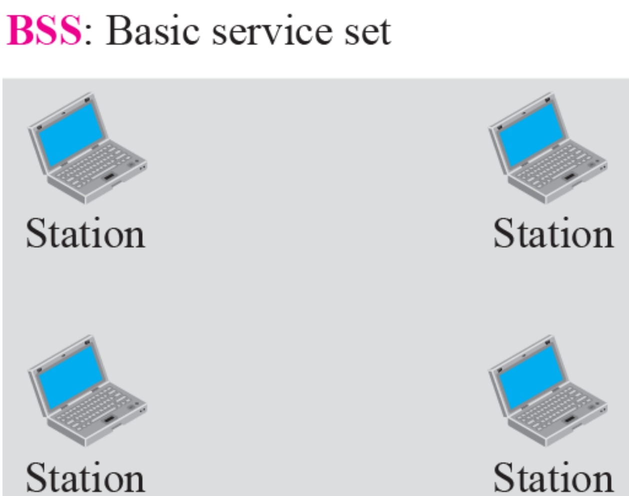
Table 3.4 *Ten-Gigabit Ethernet Implementation*

Characteristics	*10GBase-S*	*10GBase-L*	*10GBase-E*
Media	multi-mode fiber	single-mode fiber	single-mode fiber
Number of wires	2	2	2
Maximum length	300 m	10,000 m	40,000 m

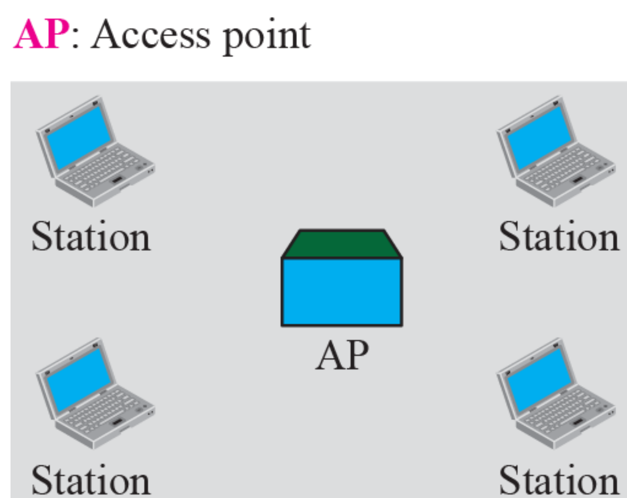
WIRELESS LANS

- **The demand for connecting devices without the use of cables is increasing everywhere.**
- **Wireless LANs can be found on college campuses, in office buildings, and in many public areas.**
- **two wireless technologies for LANs: IEEE 802.11 wireless LANs, sometimes called wireless Ethernet, and Bluetooth, a technology for small wireless LANs.**

Basic service sets (BSSs)



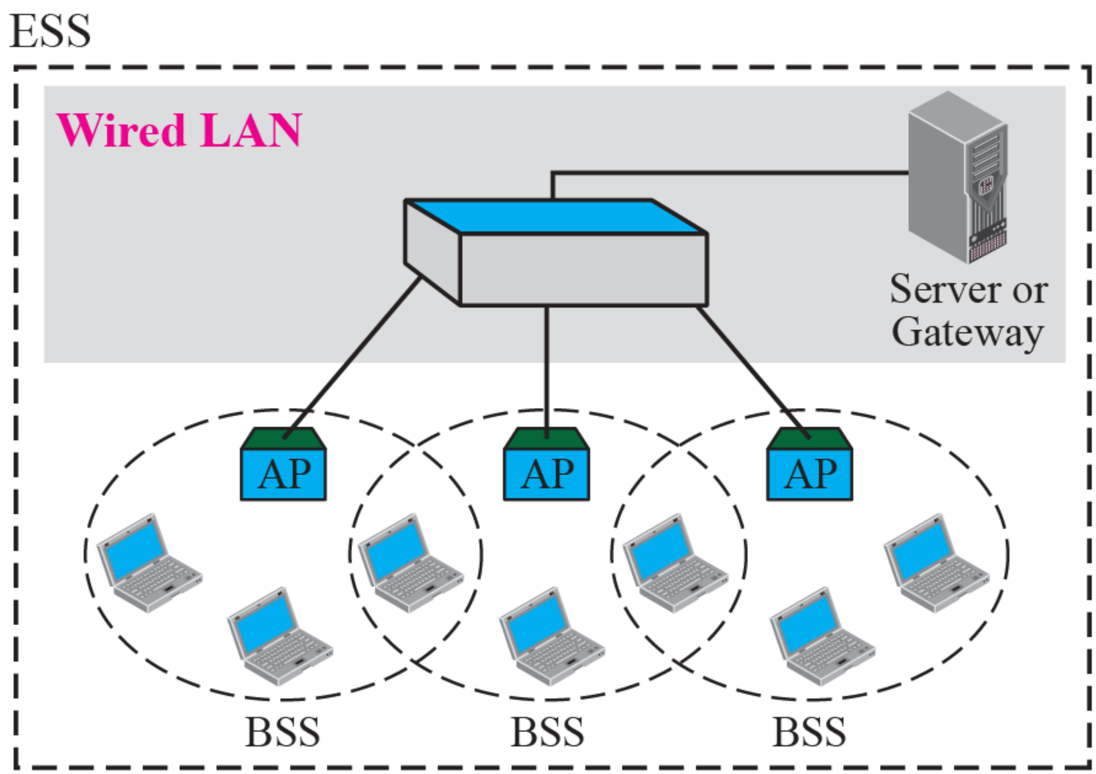
Ad hoc network (BSS without an AP)



Infrastructure (BSS with an AP)

Extended service sets (ESSs)

ESS: Extended service set
BSS: Basic service set
AP: Access point



POINT-TO-POINT WANS

- A second type of network in the Internet is the point-to-point wide area network.
- A point-to-point WAN connects two remote devices using a line available from a public network such as a telephone network.
- Some options are: traditional modem technology, DSL line, cable modem, T-lines, and SONET.

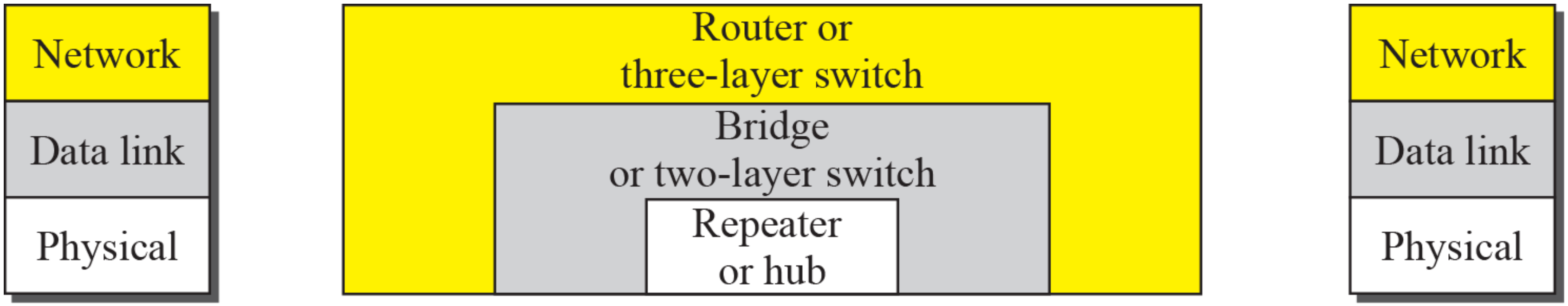
SWITCHED WANS

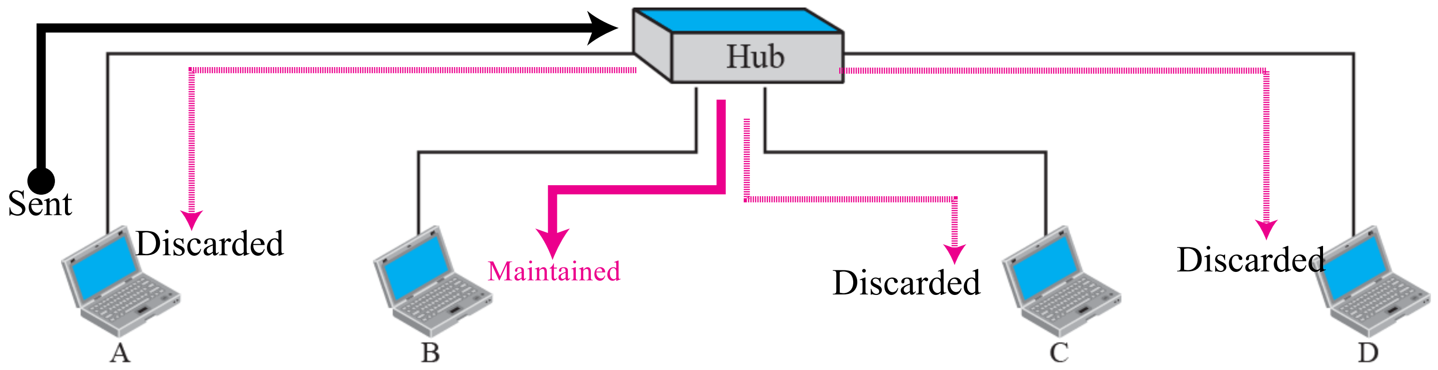
- The backbone networks in the Internet can be switched WANs.
- A switched WAN is a wide area network that covers a large area (a state or a country) and provides access at several points to the users.
- Inside the network, there is a mesh of point-to-point networks that connects switches.
- The switches, multiple port connectors, allow the connection of several inputs and outputs.
- Switched WAN technology differs from LAN technology in many ways.

CONNECTING DEVICES

- LANs or WANs do not normally operate in isolation.
- They are connected to one another or to the Internet.
- To connect LANs and WANs together connecting devices are used.
- Connecting devices can operate in different layers of the Internet model.
- Connecting devices: repeaters (or hubs), bridges (or two-layer switches), and routers (or three-layer switches).

Connecting devices

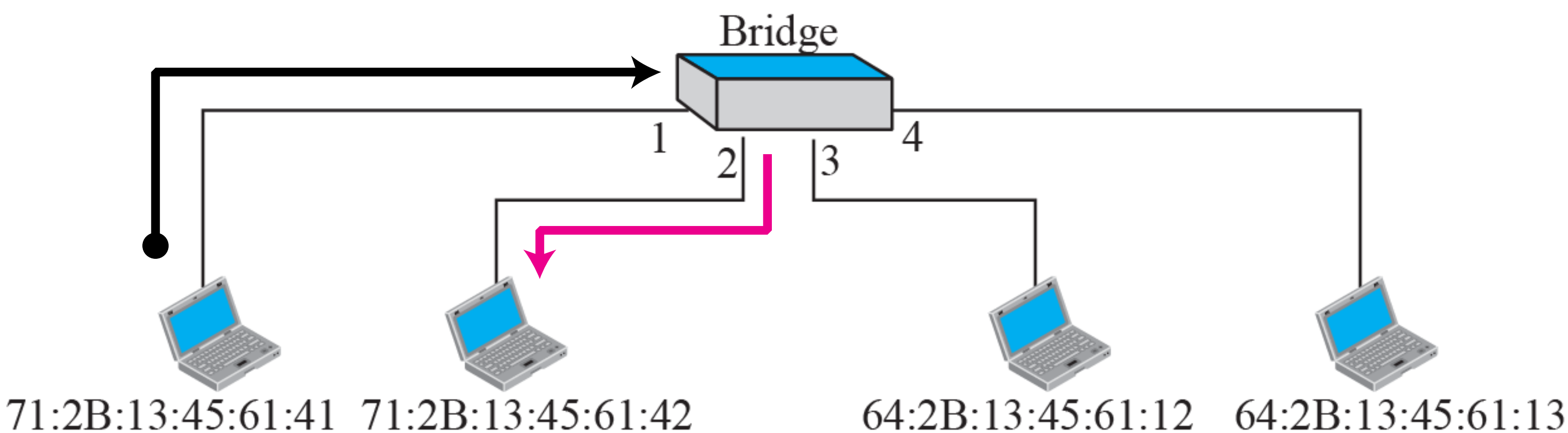




A repeater forwards every bit; it has no filtering capability.

Bridge table

Address	Port
71:2B:13:45:61:41	1
71:2B:13:45:61:42	2
64:2B:13:45:61:12	3
64:2B:13:45:61:13	4



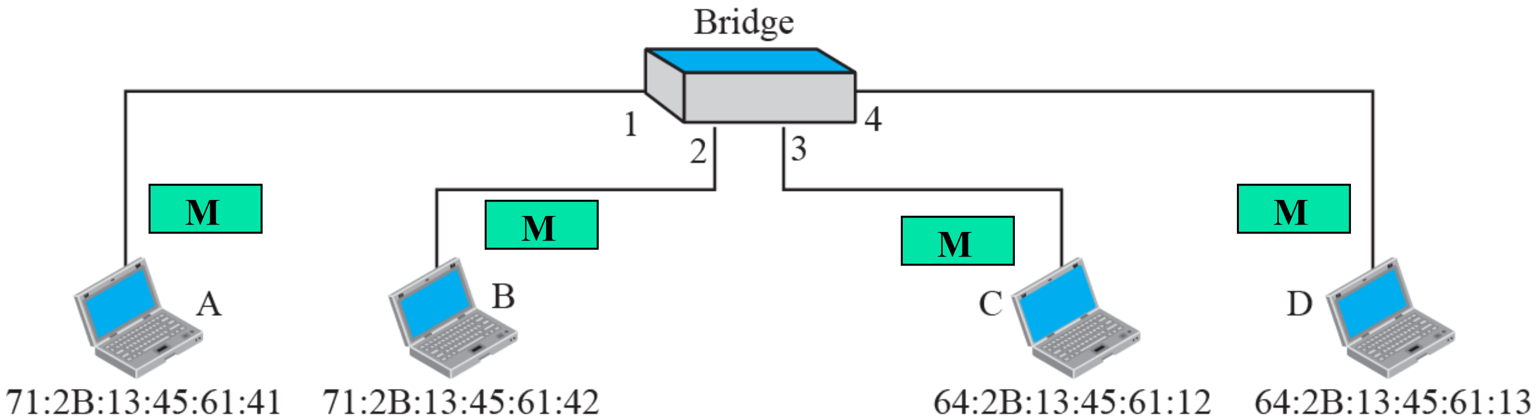
A bridge has a table used in filtering decisions.

A bridge does not change the physical (MAC) addresses in a frame.

Learning bridge – 5 min. exercise

How does a bridge build Bridge table by MAC-Learning?
What is the MAC table built after :

1) **A send a frame to D**
2) **D sends a frame to B**
3) **B sends a frame to A**
4) **C sends a frame to D**



Gradual building of Table

Address	Port

a. Original

Address	Port
71:2B:13:45:61:41	1

b. After A sends a frame to D

Address	Port
71:2B:13:45:61:41	1
64:2B:13:45:61:13	4

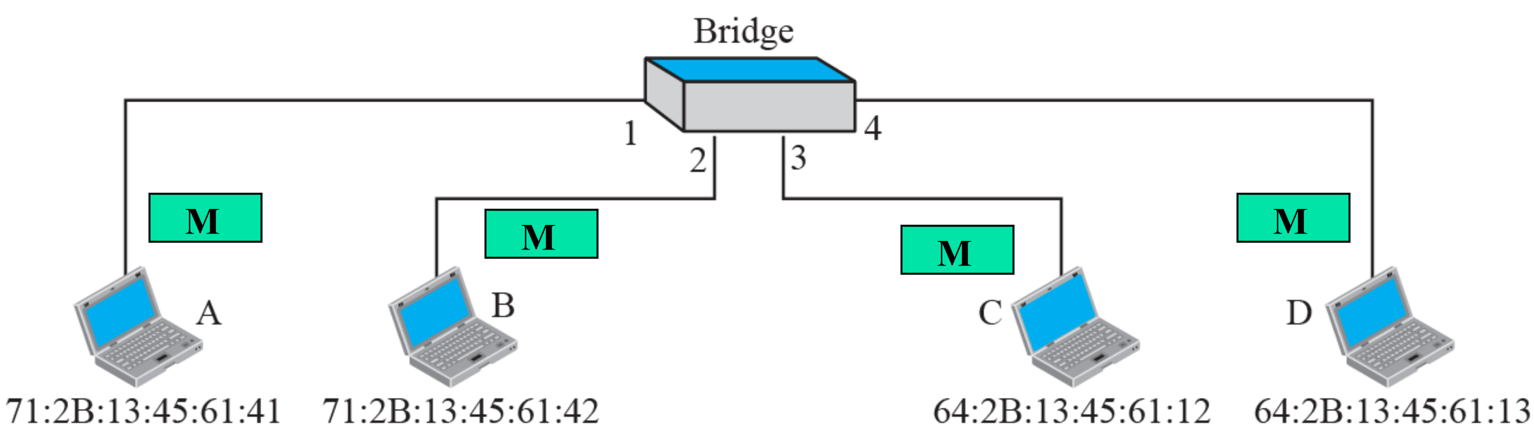
c. After D sends a frame to B

Address	Port
71:2B:13:45:61:41	1
64:2B:13:45:61:13	4
71:2B:13:45:61:42	2

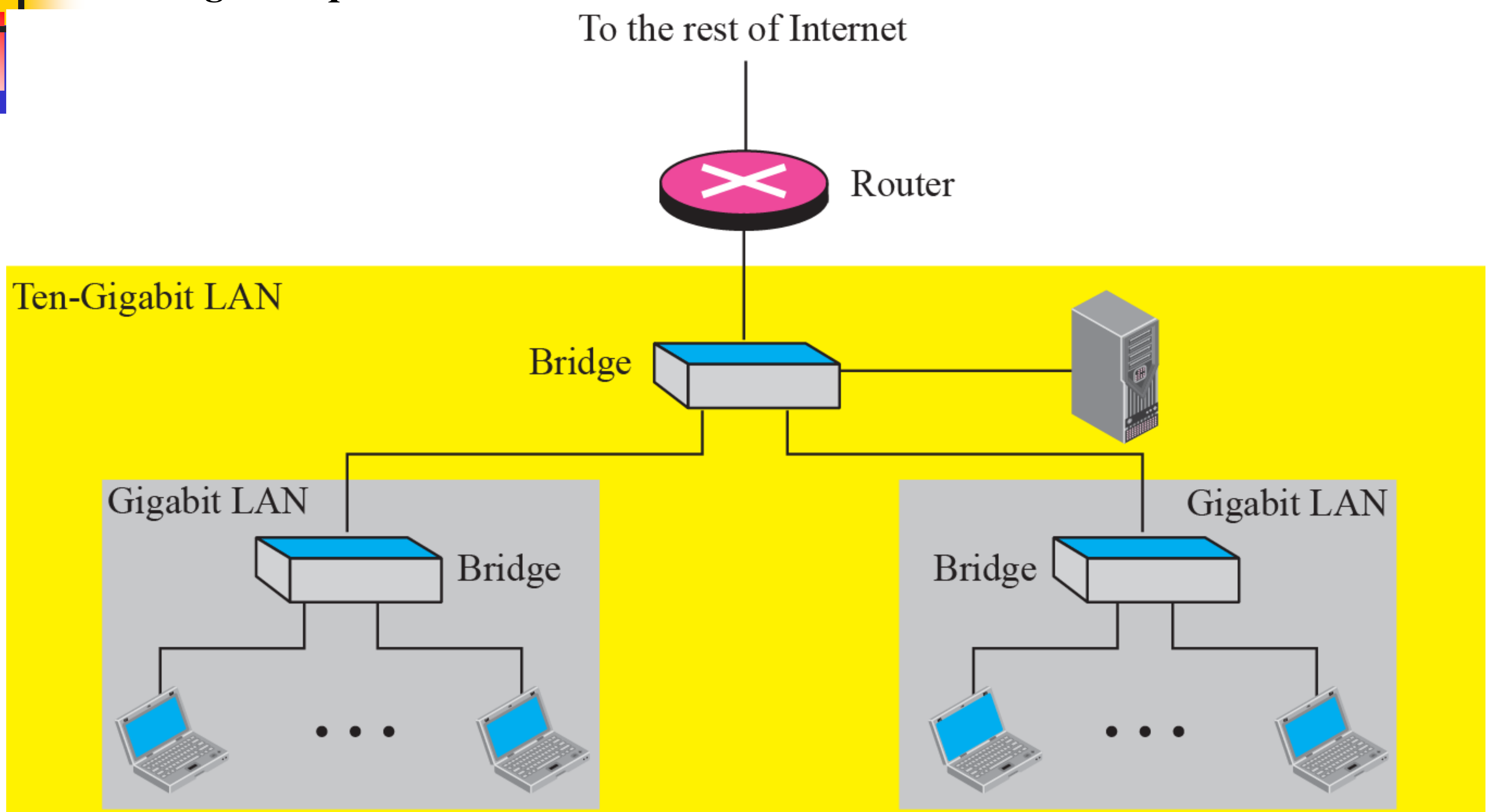
d. After B sends a frame to A

Address	Port
71:2B:13:45:61:41	1
64:2B:13:45:61:13	4
71:2B:13:45:61:42	2
64:2B:13:45:61:12	3

e. After C sends a frame to D



Routing example



A router is a three-layer (physical, data link, and network) device.

A repeater or a bridge connects segments of a LAN.
A router connects independent LANs or WANs to create an internetwork (internet).

A router changes the physical addresses in a packet.

References

TCP/IP Protocol Suite (4th Edition),

by Behrouz A. Forouzan – Used the text's slide deck

End of session